CSC 350 Assignment 5
Spring 2025

Due: Tuesday, March 25 (before class)

General Instructions

- Scan or compile snapshots of your submission into a PDF and upload it to Canvas.
- Assignments are **due before class** on the scheduled due date; generally, late assignments will not be accepted.
- Unless specified, **justify your answers always**. If justifications are missing or do not show reasonably how your answers are derived, 0 points will be assigned regardless of the correctness of the answers.
- Write **neatly** in the event you write your answers on paper. **Especially, for table completion questions, if your answers are not given neatly in well-drawn tables, the answers won't be graded**.
- **This assignment must be done by yourself without any unauthorized assistance**. Write your full name and numeric university ID on the front page of your submission, along with the following pledge statement.

I pledge my honor that all parts of this assignment were done by me individually and without any unauthorized assistance or collaboration with others.

Q1. Suppose three fair coins are flipped. (H: heads, T: tails)

(a) List the sample space.

(b) List the probabilities of every point in your sample space.

(c) What is the probability of the event that exactly two coins are tails?
Justify your answer.

(d) What is the probability that at most two coins are tails?
Justify your answer.

(e) What is the probability of the event that all coins are the same?
Justify your answer.

(f) Consider the following events:

Event A: All coins are the same
Event B: The first two coins are heads

Write the formula for the conditional probability $P(A|B)$
and calculate the probability using the formula.

Q2. Suppose a pair of fair dice are flipped.

(a) List the sample space.

(b) What is the probability that the sum of the dots is even?
Justify your answer.

(c) What is the probability that the sum of the dots is 5 or less?
Justify your answer.

(d) What is the probability that the sum of the dots is 5 or less
given that the first dice is a 2?
Justify your answer.

Q3. A student is chosen at random from a class of 80 students that has 20 honor students, 30 athletes, and 40 people that are neither honor students nor athletes.

(a) What is the probability that the student selected is an athlete, **given** that they are an honor student? Show how you calculate this probability.

(b) What is the probability that the student selected is an honor student, **given** that they are an athlete? Show how you calculate this probability.

(c) Are the events "honor student" and "athlete" independent?
Explain why.

Q4. Use Bayes' Theorem to answer this question.

A commuter crosses one of three bridges, A, B, or C, to go home from work. The commuter crosses A with probability 1/3, B with probability 1/6, and C with probability 1/2. The commuter arrives home by 6 PM 75%, 60%, and 80% of the time by crossing bridges A, B, and C, respectively. If the commuter arrives home by 6 PM, find the probabilty that bridge A was used. Also find the probabilities for bridges B and C. Show how you calculate these probabilities.

Q5. State which properities of each binary relation satisfies from the following:
reflexive, irreflexive, symmetric, antisymmetric, and transitive

(a) The "is brother of" relation on the set of people.

(b) $R = \{(x, y) \mid x \; divides \; y\}$ over natural numbers

(c) The subset relation on sets.

Q6. Given the three relations "less," "greater," and "notEqual" over the natural numbers N, find each of the following compositions.

(a) less $\circ$ greater.

(b) greater $\circ$ notEqual.

(c) notEqual $\circ$ less.

Q7. Find the closure of each of the following relations over the set $A = \{a, b, c, d\}$.

(a) Find $s(R)$ for $R = \{(a, b), (b, c), (c, d)\}$.

(b) Find $t(R)$ for $R = \{(a, b), (b, c), (c, d), (d, a)\}$.

(c) Find $r(R)$ for $R = \{(a, c), (b, d)\}$.

(d) Find $st(R)$ for $R = \{(a, b), (b, c), (c, d)\}$.

Q8. Suppose G is the following weighted digraph, where the triple (i, j, d) represents edge (i, j) with distance d:

$$G = \{(1, 2, 20), (1, 4, 5), (2, 3, 10), (3, 4, 10), (4, 3, 5), (4, 2, 10)\}.$$

(a) Draw the weighted digraph representing G.

(b) Produce the weighted adjacency matrix for G.

(c) Use Floyd's algorithm for finding the **shortest path** (4.1.9) to compute the two matrices M and P, representing the shortest distances and the shortest paths in G.